

Mustafa Shoukat

Senior AI Engineer · Enterprise AI Architecture & Technical Leadership

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Prepared for: Staff AI Engineer — Teradata

PROFILE

Mustafa Shoukat is an AI engineer operating at **Staff-level scope** — he owns architecture end-to-end for distributed, fault-tolerant AI platforms, not just individual features. He has architected and shipped **20+ production systems** for enterprise and government-grade clients across **Saudi Arabia, the UAE, the United States, and the United Kingdom**, consistently taking AI from prototype to hardened, observable, day-to-day production inside regulated environments.

His signature work spans two of the hardest problems in enterprise AI: **trustworthy retrieval over sensitive data** (a multi-tenant, air-gapped RAG platform with per-organization isolation and 5-level access control) and **measurable model quality** (a Claude-as-judge evaluation-and-fine-tuning pipeline that took an Arabic LLM to **#1 on the AraGen Leaderboard**).

CAREER TRAJECTORY

A clear progression from building models to owning platforms:

- **Data Scientist** (construction-tech, US) — ML/NLP and document-processing pipelines hardened against messy real-world data.
- **Generative AI Consultant** (Arectec, US) — a production Text-to-SQL RAG application letting non-technical users query enterprise databases in plain English.
- **Agentic AI Engineer** (Watad / Navid, Riyadh) — end-to-end owner of **ByanRAG 2.2**, a production Arabic-first, multi-tenant RAG platform for regulated networks.
- **Independent AI Software Engineer** (remote; KSA / UAE / UK / US) — architects and ships production AI systems and advises organizations moving AI from pilot to operations.

The arc is individual contributor → system owner → **architect and technical lead**.

MOST RELEVANT ENTERPRISE-AI ACHIEVEMENTS

- **Owned architecture end-to-end for ByanRAG 2.2** — a multi-tenant, Arabic-first RAG platform with org-isolated vector collections, a 5-level Casbin RBAC hierarchy enforced on every endpoint, a 14-panel admin console, full audit logging, and **on-premises / air-gapped deployment** for government networks.
- **Designed the hybrid retrieval pipeline** — dense embeddings + BM25 sparse search + Reciprocal Rank Fusion + CrossEncoder reranking, with NLI faithfulness scoring — powering source-cited, bilingual streaming chat.
- **Built a Claude-as-judge reward pipeline** (3C3H metric) and NLI-based evaluation guardrails that took **Yehia R1 to #1 on the AraGen Leaderboard**.
- **Delivered an agentic government-contracting platform** — LangGraph multi-agent workflows over SAM.gov solicitations for automated requirement analysis, bid/no-bid decisions, and PDF proposal generation.
- **Set engineering standards** — a 600+ test suite, CI/CD, Docker Compose, nginx/TLS, and full Prometheus / Grafana / Loki observability.

TECHNICAL LEADERSHIP

Beyond delivery, Mustafa sets the standard others build to: he defined testing, deployment, and model-integration practices across his team, **mentored engineers**, and has **trained 150+ professionals** in data science, ML, and AI engineering. He is an active open-source contributor with merged pull requests to **LangChain** (92k★), **RAGAS** (7k★), and **Unstructured** (14k★).

CORE STACK

Python · Java · FastAPI · LangGraph / CrewAI / MCP · Qdrant / Milvus / FAISS / pgvector · PostgreSQL / MongoDB · Celery / Redis · Docker · AWS / GCP / Azure · Prometheus / Grafana / Loki